

Real-Time Leak Detection & AI-Dispatched Maintenance

The Problem

- **Invisible water loss**

A single stuck valve in a high-footfall facility can waste thousands of litres before anyone notices.

- **Fixed cleaning schedules**

Housekeeping runs on a timer, not actual usage — over-cleaning low-traffic units and under-cleaning busy ones.

- **Reactive maintenance**

Facility teams find out about failures from complaints, not from the sensors already reporting them.

Our Approach

- **Continuous rule-based detection**

Purpose-built rules catch leaks, hygiene gaps, and sensor faults deterministically and auditably.

- **AI diagnosis on every alert**

An LLM converts each raw trigger into a prioritized, human-readable ticket with cause and recommended action.

- **Resilient by design**

A deterministic fallback keeps dispatch running even if the AI service is unreachable.

DETECT

3 failure classes caught in real time

DIAGNOSE

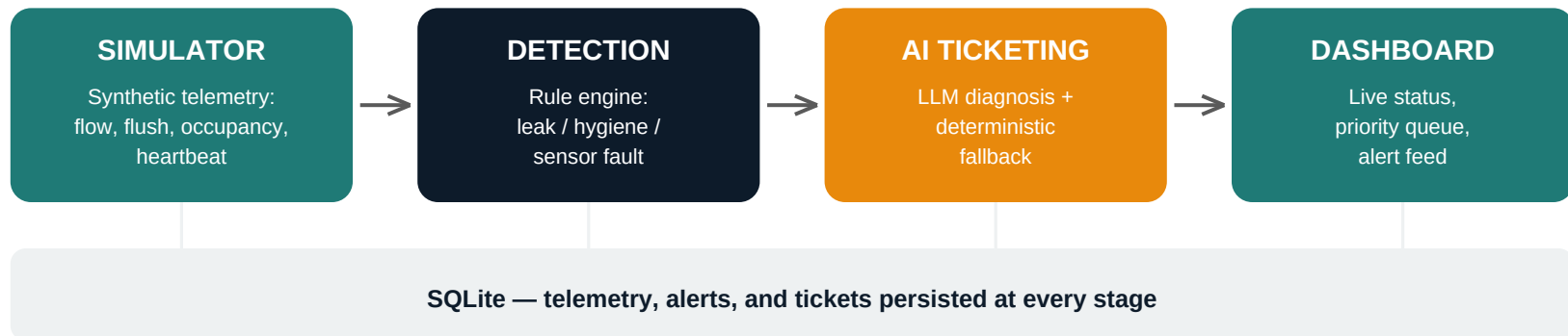
AI turns raw alerts into technician-ready tickets

DISPATCH

Fails safe — never goes silent

Demonstration scenario: an airport terminal wing with 4 monitored restroom units, ~40,000 passengers/day

From Sensor to Dispatched Ticket



Design Highlights

- Deterministic detection rules keep the system auditable — every alert traces to a fixed threshold, not a model's judgment call.
- The AI layer sits after detection, not instead of it — it explains and prioritizes what the rules already caught with certainty.
- Every ticket records generated_by (claude-llm or rules-fallback), so the dispatch pipeline never has an unexplained gap.

WHY IT'S RESILIENT

If the Anthropic API is unreachable, the same JSON shape is produced by a rules-based generator — the dashboard, the queue, and the technician workflow never know the difference.

Built for Speed, Reasoned for Trust

Technology Stack

Python 3

Core language across simulation, detection, and dispatch logic

SQLite

Zero-setup persistence for telemetry, alerts, and tickets

Streamlit

Live operations dashboard — status grid, charts, ticket queue

Anthropic API

Claude Sonnet 4.6 for structured ticket diagnosis

How AI Is Used

- **Structured diagnosis, not chat**

Every response is constrained to a strict JSON schema so it drops straight into the ticket queue with zero manual interpretation.

- **Context-grounded reasoning**

The model is given real facility context (footfall, available teams) so priority scores reflect operational reality.

- **Fails safe, never silent**

A deterministic generator mirrors the AI's output format if the API is unavailable — dispatch never stops.

- **Auditable by design**

Every ticket is tagged with which path produced it, so the AI's contribution is always visible and verifiable.

SAMPLE AI-GENERATED TICKET

```
{"title": "Continuous flow detected at R101", "priority_score": 90, "priority_label": "CRITICAL",  
  "likely_cause": "Probable stuck-open flush valve causing uninterrupted flow.",  
  "recommended_action": "Isolate supply line and inspect flush valve diaphragm.",  
  "assigned_team": "PLUMBING", "est_water_loss_lpd": 3369.6}
```

Why This Approach Matters

3

Failure classes detected
automatically

~5 min

Average time from leak
onset to ticket dispatch

100%

Dispatch uptime, even if
the AI service is down

L/day

Water loss quantified
per open leak ticket

Business & Sustainability Alignment

- **Water conservation**

Every leak ticket carries a projected litres-per-day figure, turning silent waste into a tracked, actionable metric — directly supporting KOHLER's conservation commitment.

- **Operational efficiency**

Usage-based hygiene alerts replace fixed cleaning schedules, directing housekeeping effort where footfall actually demands it.

- **Deployment feasibility**

The architecture maps directly onto real infrastructure: simulated sensors become real IoT telemetry, and the same detection/dispatch pipeline runs unchanged against a live MQTT or REST feed.

From reactive maintenance to predictive, sustainable facility operations.